Name: _____ Register No: _____

Experiment 5

Question: Data Collection and Initial Exploration.

Code

```
import json
import re
from collections import Counter

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

def exp_5():
    # Data collection: here we build it; in practice use read_csv / read_excel /
    # read_json / read_sql / web scraping / APIs
    rng = np.random.default_rng(7)
    n = 150
    df = pd.DataFrame({
        "StudentID": range(1, n + 1),
        "Gender": rng.choice(["M", "F"], n),
        "Math": rng.integers(35, 100, n),
        "Science": rng.integers(30, 100, n),
        "Attendance": rng.normal(80, 10, n).round(1),
    })
    df.to_csv("collected_data.csv", index=False)
    df = pd.read_csv("collected_data.csv")

    # Initial exploration
    print("Shape         :", df.shape)
    print("Columns       :", list(df.columns))
    print("Data types    :\n", df.dtypes, "\n")
    print("First 5 rows  :\n", df.head(), "\n")
    print("Last 5 rows   :\n", df.tail(), "\n")
    df.info()
    print("\nSummary stats :\n", df.describe(), "\n")
    print("Missing values:\n", df.isnull().sum(), "\n")
    print("Duplicates    :", df.duplicated().sum())
    print("Gender counts :\n", df["Gender"].value_counts())
    print("Correlation   :\n", df[["Math", "Science", "Attendance"]].corr())

    sns.pairplot(df[["Math", "Science", "Attendance", "Gender"]], hue="Gender")
    plt.show()


exp_5()
```

Output

```
Shape         : (150, 5)
Columns       : ['StudentID', 'Gender', 'Math', 'Science', 'Attendance']
Data types    :
 StudentID       int64
Gender            str
Math            int64
Science         int64
Attendance    float64
dtype: object

First 5 rows  :
    StudentID Gender  Math  Science  Attendance
0          1      F    89       82        63.6
1          2      F    89       47        85.1
2          3      F    66       85        79.4
3          4      F    96       86        84.1
4          5      F    74       77        70.1

Last 5 rows   :
      StudentID Gender  Math  Science  Attendance
145        146      F    62       70        65.9
146        147      M    45       57        88.5
147        148      F    68       74        87.1
148        149      F    70       48        78.5
```

```
149        150      M    96       72        62.9

<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   StudentID   150 non-null    int64
 1   Gender      150 non-null    str
 2   Math        150 non-null    int64
 3   Science     150 non-null    int64
 4   Attendance  150 non-null    float64
dtypes: float64(1), int64(3), str(1)
memory usage: 6.0 KB

Summary stats :
         StudentID        Math     Science  Attendance
count  150.000000  150.000000  150.000000  150.000000
mean    75.500000   69.500000   64.653333   79.497333
std     43.445368   19.584184   20.293811    9.620706
min      1.000000   35.000000   30.000000   47.500000
25%     38.250000   53.000000   47.000000   73.775000
50%     75.500000   70.000000   68.000000   79.850000
75%    112.750000   88.500000   82.000000   85.100000
max    150.000000   99.000000   98.000000  100.300000

Missing values:
 StudentID     0
Gender        0
Math          0
Science       0
Attendance    0
dtype: int64

Duplicates    : 0
Gender counts :
 Gender
F    78
M    72
Name: count, dtype: int64
Correlation   :
                Math   Science  Attendance
Math        1.000000 -0.164341   -0.034349
Science    -0.164341  1.000000    0.040950
Attendance -0.034349  0.040950    1.000000
```

Graph

